Classification or Regression?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Recall and choose your starting point

> Record your starting explanation for Classification or Regression? before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Features, Labels, Targets

Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Practise with fading help

> Complete the supported Classification or Regression? trace and record the first step you can explain without help.

Use W2: Trace both checks for **Classification** and **regression**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Classification or Regression? explanation against the supplied facts before discussing it.

Use the worked case for **Classification** and **regression**. Proposed explanation: “Numbers always mean **regression**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Compare new evidence independently

For three fictional print jobs, actual completion times are 4, 5 and 9 minutes; estimates are 6, 5 and 8 minutes in matching order. Slow means more than 5 minutes.

Write actual and predicted category lists, count correct categories, calculate each **absolute error** and the mean error in minutes. State what each result measures.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Classification or Regression? target to a trace and a justified conclusion.

System A predicts **class** codes 0=oak,1=maple. System B predicts leaf area in cm². Name the task types and explain why code 1 is not 'one more species' than code 0.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Classification or Regression? case and show the deciding evidence.

Task A predicts healthy/spotted; B predicts next week's height in cm.

Choose task and **output** types.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

> Task A predicts whether a plant is fern or grass. Task B predicts its height in centimetres. Which is **classification** and which is **regression**?

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 3 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Classification or Regression? adds to Responsible Plant Classifier and keep the evidence.

Confirm whether Plant **Class**ifier predicts a health category or a numerical severity score; document why.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

System A predicts **class** codes 0=oak,1=maple. System B predicts leaf area in cm². Name the task types and explain why code 1 is not 'one more species' than code 0.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Classify the target meaning

> Classify the target meaning. Apply the supplied Classification or Regression? facts and keep your reasoning.

At dispatch, identify the task for predicting route type coded 1 or 2, duration in whole minutes, and whether duration exceeds 10 minutes. Explain your choices.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace both checks

> Trace both checks. Apply the supplied Classification or Regression? facts and keep your reasoning.

For actual 14 minutes and estimate **12 minutes**, calculate **absolute error** and both slow categories using greater than 10. Explain why a correct category does not mean an exact duration.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Repair a category average

> Repair a category average. Apply the supplied Classification or Regression? facts and keep your reasoning.

Indoor is coded 1 and outdoor is coded 2. Someone averages **prediction**s and reports route type 1.5 as an accurate halfway category. Explain the problem and name an appropriate check.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Compare new evidence independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Compare new evidence independently. Apply the supplied Classification or Regression? facts and keep your reasoning.

For three fictional print jobs, actual completion times are 4, 5 and 9 minutes; estimates are 6, 5 and 8 minutes in matching order. Slow means more than 5 minutes.

Write actual and predicted category lists, count correct categories, calculate each **absolute error** and the mean error in minutes. State what each result measures.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Specify a useful output

> Specify a useful output. Apply the supplied Classification or Regression? facts and keep your

> reasoning.

A user needs the arrival duration in minutes, but a report gives only slow or not slow. What task should be specified, and what information was lost by the category **output**?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Classification or Regression?

> Distinguish predicting growth height from predicting a growth category. Explain the labelled step before checking the changed condition.

Distinguish predicting growth height from predicting a growth category.

Actual heights are [4, 7, 9] cm; estimates are [5, 6, 10] cm. Tall means strictly greater than 6 cm.

All **input**s and outcomes in this worked example are invented **class**room data; no real model run is claimed.

AI ACADEMY / AI-4/3

Actual and estimated heights (paired order)



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/3

Classification or Regression?

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Regression errors

Absolute errors are 1, 1, 1 cm; mean absolute error is 1 cm.



2. Actual categories

Actual classes are short, tall, tall.



3. Predicted categories

Estimated classes are short, short, tall: 2/3 correct.

CHANGE ONE THING

Change the category boundary to strictly greater than 7 cm.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Did **regression** MAE improve when the boundary changed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant **Class**ifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant **Class**ifier

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For your Responsible Plant **Class**ifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Distinguish predicting growth height from predicting a growth category.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Estimated **class**es are short, short, tall: 2/3 correct.

1. Actual categories

2. **Regression** errors

3. Predicted categories

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Predicted categories Estimated **class**es are short, short, tall: 2/3 correct. Read the picture in order. Estimated **class**es are short, short, tall: 2/3 correct.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change the category boundary to strictly greater than 7 cm. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Actual and estimated categories both become short, short, tall: 3/3 correct.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Actual and estimated categories both become short, short, tall: 3/3 correct. The numerical estimates did not improve; the decision definition changed.

Repair a missing building block

Review Features, Labels, Targets (ai-4/2). Goal: Separate **prediction input**s from observed targets. Short practice: Why can a historical table contain a field that is an invalid **input**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Classification or Regression? picture from its labels, then record a relevant limit.

01. Classification or Regression?: the case

AI ACADEMY / AI-4/3 · 01 · LOOK AT THE FACTS

Classification or Regression?: the case

The situation

Distinguish predicting growth height from predicting a growth category.

Supplied evidence

Actual heights are [4, 7, 9] cm; estimates are [5, 6, 10] cm. Tall means strictly greater than 6 cm.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain classification or regression? using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

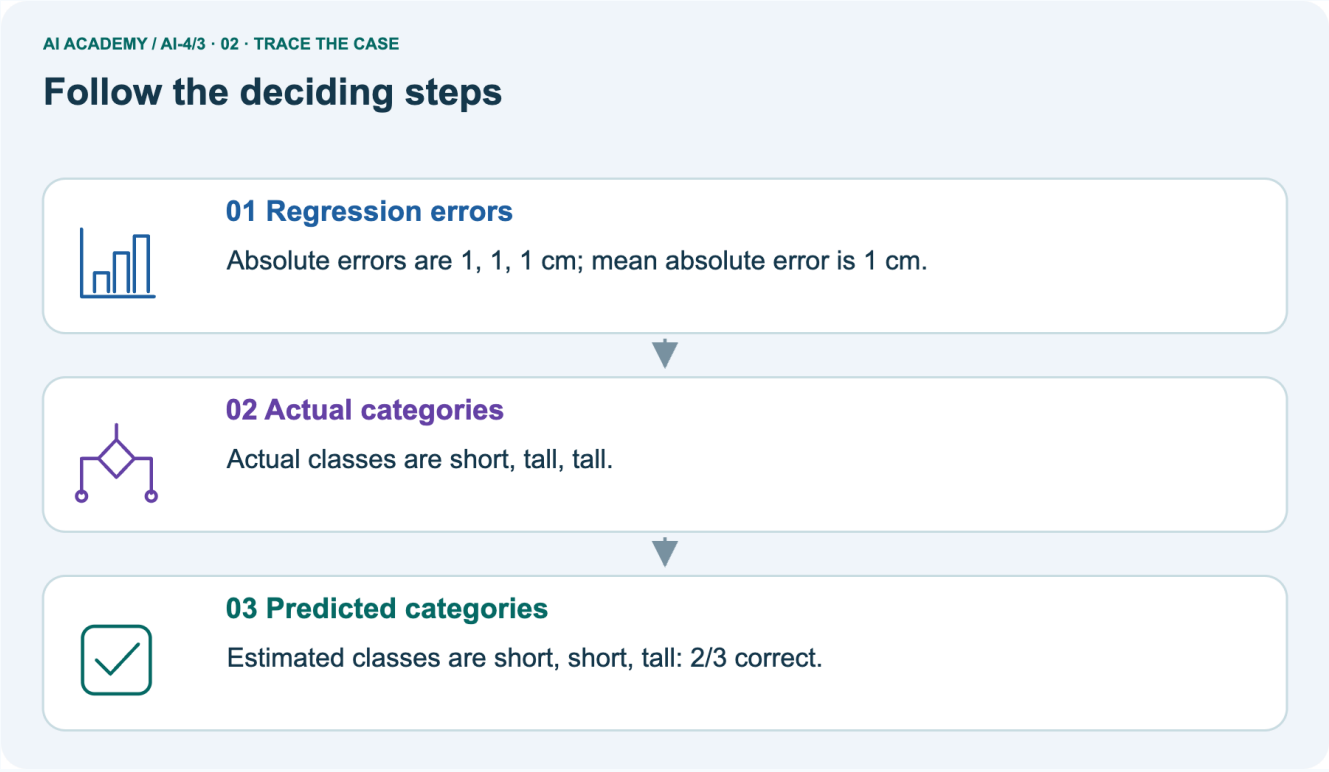
Distinguish predicting growth height from predicting a growth category. Supplied facts: Actual heights are [4, 7, 9] cm; estimates are [5, 6, 10] cm. Tall means strictly greater than 6 cm.

All **input**s and outcomes in this worked example are invented **class**room data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Regression errors: **Absolute error**s are 1, 1, 1 cm; mean **absolute error** is 1 cm. → Actual categories: Actual **class**es are short, tall, tall. → Predicted categories: Estimated **class**es are short, short, tall: 2/3 correct.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Actual and estimated heights (paired order)



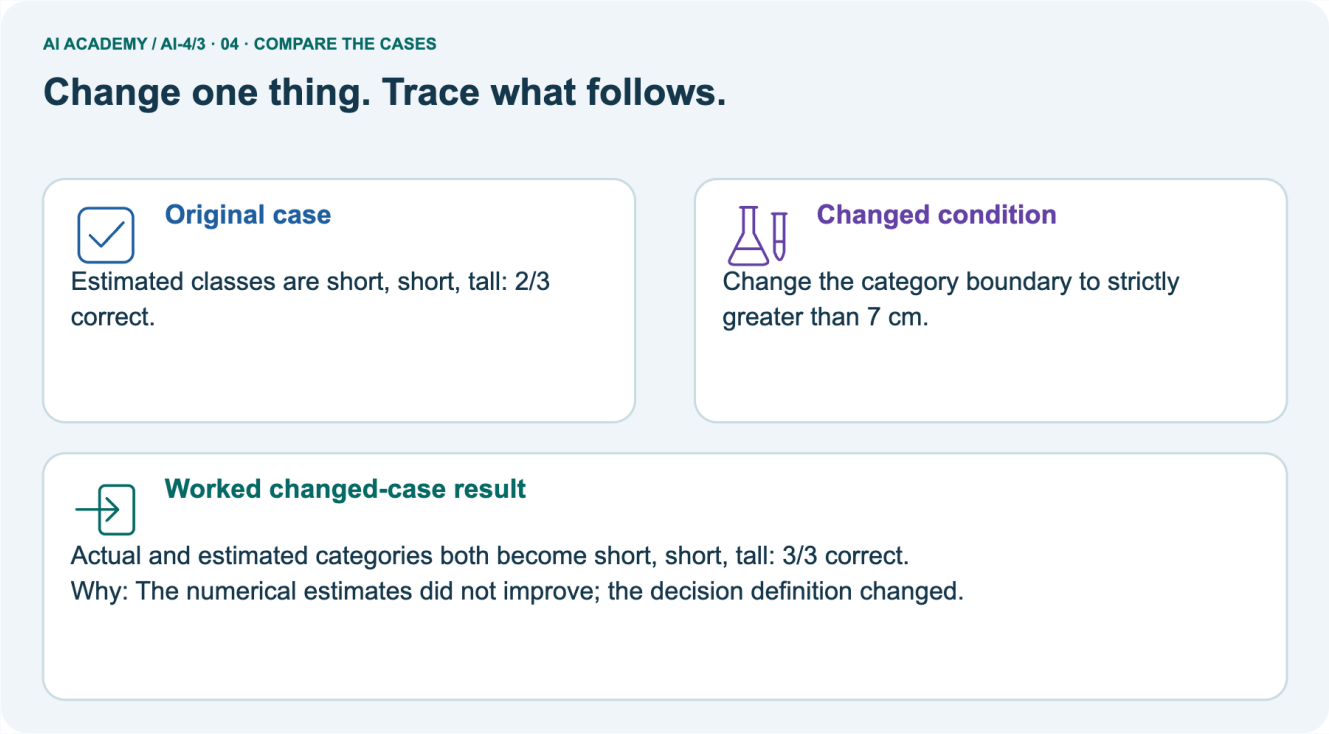
Follow the labelled picture with the matching explanation. Open the vector diagram

Actual and estimated heights (paired order). Units: cm. actual1: 4. estimate1: 5. actual2: 7. estimate2: 6. actual3: 9. estimate3: 10.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

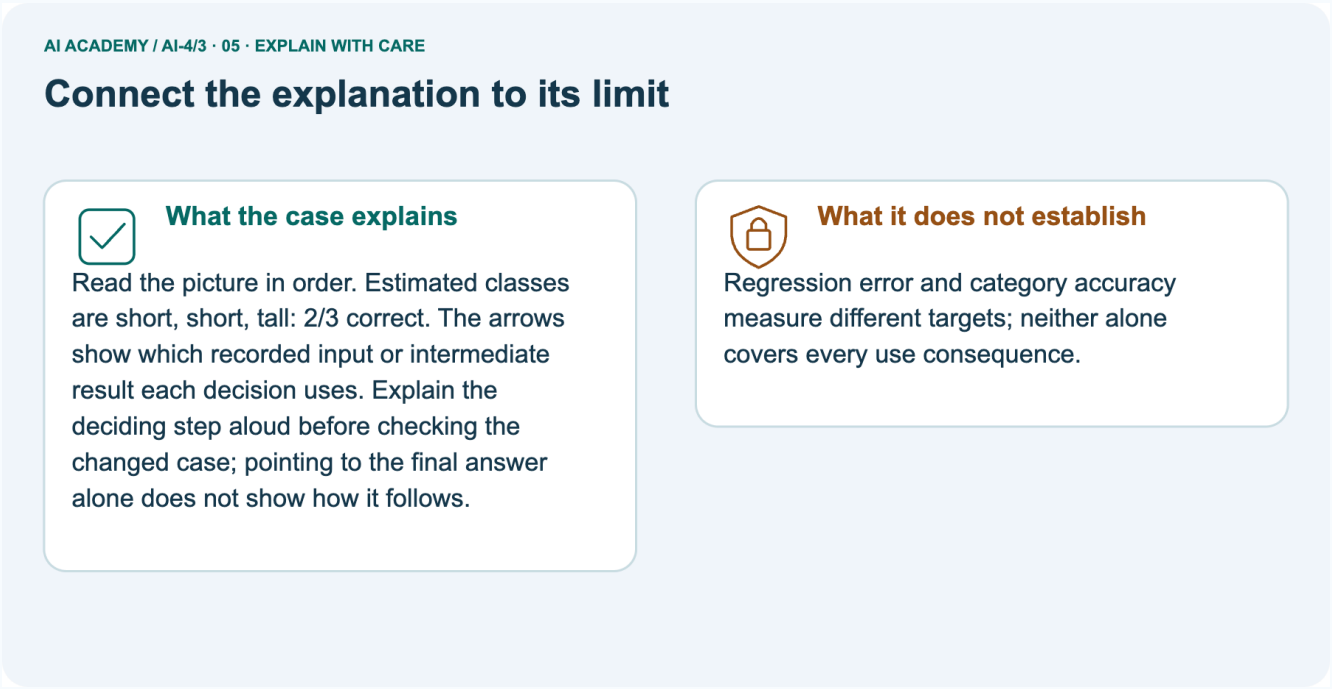
Original case: Estimated **class**es are short, short, tall: 2/3 correct. Changed condition: Change the category boundary to strictly greater than 7 cm.

Expected result: Actual and estimated categories both become short, short, tall: 3/3 correct. Reason: The numerical estimates did not improve; the decision definition changed.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Estimated **class**es are short, short, tall: 2/3 correct. The arrows show which recorded **input** or intermediate result each decision uses.

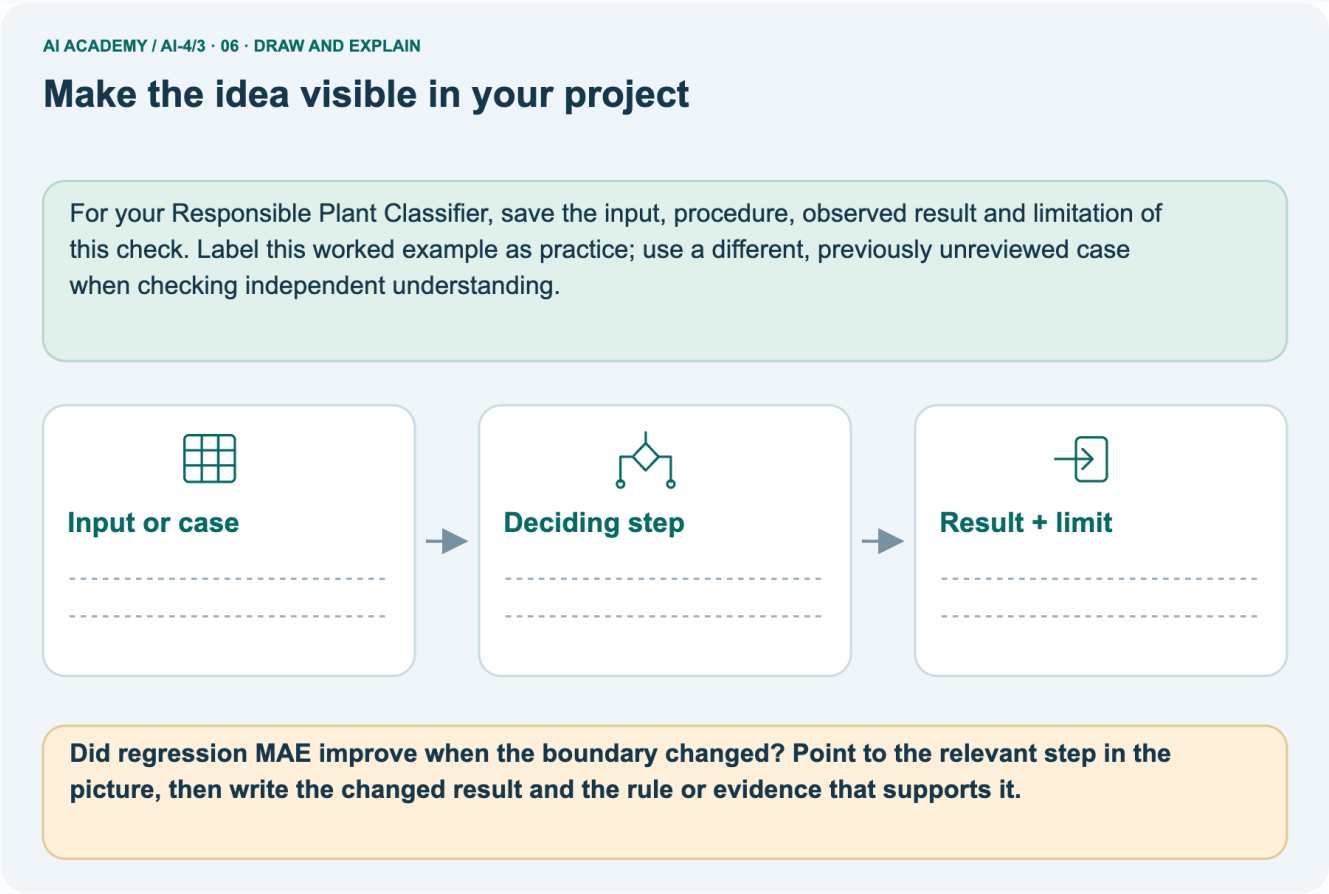
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: **Regression** error and category accuracy measure different targets; neither alone covers every use consequence.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant **Class**ifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Did **regression** MAE improve when the boundary changed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.